Tutorial 3: Bayesian Estimation and Regularization

Three required tasks and one optional challenge

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How to use this tutorial

- Attempt each problem before reading its solution.
- Write the main calculation or argument clearly.
- Explain what the result means, not only the final number or formula.
- Complete Tasks 1–3 first. The final task is an optional challenge.

Task 1: Beta updating

We observe four Bernoulli trials:

$$m = 3, \qquad n = 4.$$

Consider two priors:

$$\text{Prior A: } \mathrm{Beta}(2, 2), \qquad \text{Prior B: } \mathrm{Beta}(20, 20).$$

Both priors have mean 0.5, but Prior B is much more concentrated.

1. Find the posterior distribution under each prior.
2. Compute the posterior mean and next-trial success probability for each prior. Compare them with the MLE.
3. For Prior A, compute the MAP estimate. Explain why it differs from the posterior mean.

Task 2: Gaussian updating

Suppose our prior belief about an unknown mean is

$$\mu \sim \mathcal{N}(10, 4).$$

We observe $n = 2$ measurements with

$$\bar{x} = 14, \qquad \sigma^2 = 8,$$

where σ^2 is the known observation variance.

1. Compute the prior precision and data precision. Which source receives more weight?
2. Compute the posterior mean and posterior variance.
3. Let x_* be a new observation from the same population. Find its predictive mean and variance, and explain why the predictive variance is larger.

Task 3: MAP and ridge

In the lecture, ridge regression used a zero-centered Gaussian prior. Now suppose instead

$$\boldsymbol{w} \sim \mathcal{N}(\boldsymbol{w}_0, \tau^2 \boldsymbol{I}),$$

while the regression model is

$$\boldsymbol{y} \mid \boldsymbol{w} \sim \mathcal{N}(\boldsymbol{X}\boldsymbol{w}, \sigma^2 \boldsymbol{I}).$$

1. Derive the MAP objective up to constants. What replaces the usual $\|\boldsymbol{w}\|_2^2$ penalty?
2. Differentiate the objective and derive the normal equations.
3. Explain what happens when $\boldsymbol{w}_0 = \boldsymbol{0}$ and when $\tau^2 \to \infty$.

Optional: Ridge experiment

Write a Python program to study how ridge regression behaves when two features are highly correlated. Generate a synthetic dataset with

$$n = 80, \qquad \boldsymbol{w}_{\text{true}} = (2, 2)^\top.$$

Generate x_1 from a standard Gaussian distribution and construct

$$x_2 = x_1 + \text{small Gaussian noise},$$

so that the two features are almost identical. Generate the target using

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{w}_{\text{true}} + \text{Gaussian noise}.$$

1. Implement ordinary least squares and ridge regression with $\lambda = 1$.
2. Compare their fitted coefficients and training MSE.
3. Repeat the experiment 100 times with new response noise and compare the variability of the fitted coefficients.
4. Explain why ridge produces more stable coefficients.

Tutorial 3 complete

Questions and discussion